

Week 8 Workshop Solutions

Question 1

A white noise process u_t is defined as a process satisfying $E(u_t) = 0$, $Var(u_t) = \sigma^2$ and $Corr(u_t, u_{t-k}) = 0$ for $k = 1, 2, \dots$

That is, a white noise process has zero mean, constant variance, and no serial correlation.

An MDS satisfies

$$E(u_t | F_{t-1}) = 0.$$

White noise focuses on unconditional properties while a martingale difference sequence is a conditional mean restriction.

Question 2

(i) Using LIE

$$\begin{aligned} E(u_t) &= E(E(u_t | F_{t-1})) \\ &= E(0) = 0. \end{aligned}$$

and so the unconditional mean of u_t is zero.

(ii) Let $k \geq 1$. Since $E(u_t) = E(u_{t-k}) = 0$ and using LIE we have

$$\begin{aligned} Cov(u_t, u_{t-k}) &= E(u_t - E(u_t))(u_{t-k} - E(u_{t-k})) \\ &= E(u_t u_{t-k}) \\ &= E(E(u_t u_{t-k} | F_{t-1})) \\ &= E(u_{t-k} E(u_t | F_{t-1})) \\ &= E(u_{t-k} \times 0) \\ &= 0 \end{aligned}$$

Therefore u_t is uncorrelated with its past values and hence is not autocorrelated.

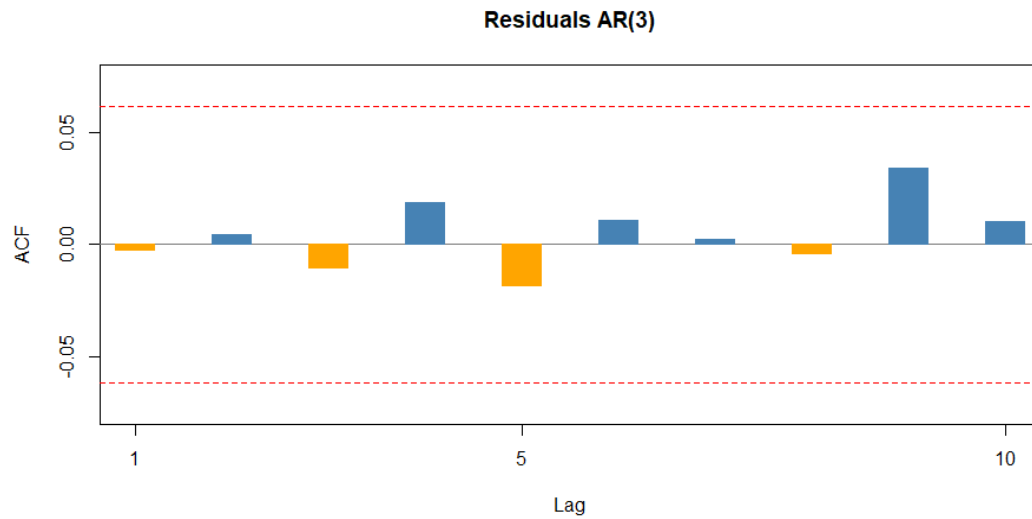
Question 3

The AR(3) model would be selected because it has the lowest AIC.

```
> aic_table
  Model      AIC
1 AR(1) 2992.369
2 AR(2) 2905.579
3 AR(3) 2847.132
4 AR(4) 2849.065
5 AR(5) 2850.980

> aic_table[order(aic_table$AIC), ]
  Model      AIC
3 AR(3) 2847.132
4 AR(4) 2849.065
5 AR(5) 2850.980
2 AR(2) 2905.579
1 AR(1) 2992.369
```

The following ACF plot of the AR(3) residuals shows that all autocorrelations lie within the approximate 95% confidence bounds. Therefore, we do not find strong evidence of serial correlation remaining in the residuals. This suggests that the AR(3) model is reasonably adequate for modelling the dependence in the series.



In practice (applied work), it is not always easy to obtain a residual ACF plot where **every** autocorrelation lies inside the dotted confidence bands especially when we look at many lags.